IV. Principal Component Analysis (PCA)

Exploratory Data Analysis & Unsupervised Learning



AIIS
Department of Applied Mathematics and Statistics

Content

1 Motivation

2 Principal Directions (PDs) & Components (PCs)

3 Dimensional Reduction Reconstruction

4 Correlation Circle

5 Contribution Scores

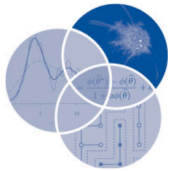
6 Variable Clustering

Resources

- Principal Component Analysis by Hervé Abdi and Lynne J. Williams.

Overview

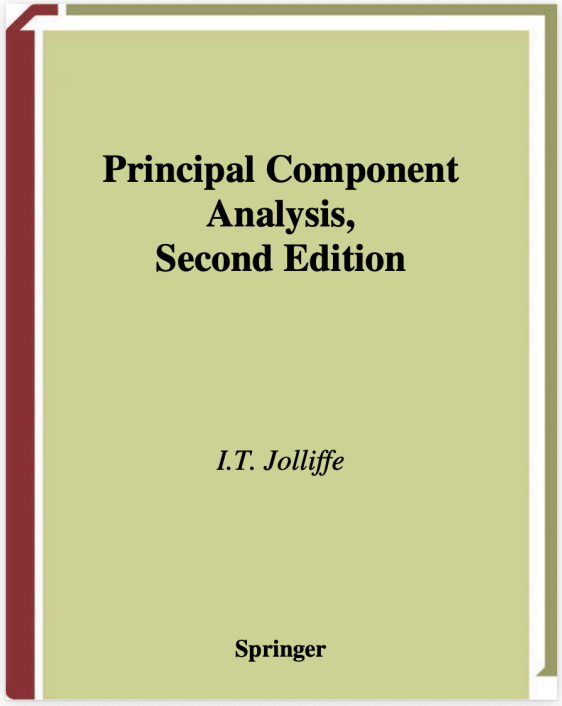
Principal component analysis



Hervé Abdi[1]* and Lynne J. Williams[2]

Principal component analysis (PCA) is a multivariate technique that analyzes a data table in which observations are described by several inter-correlated quantitative dependent variables. Its goal is to extract the important information from the table, to represent it as a set of new orthogonal variables called principal components, and to display the pattern of similarity of the observations and of the variables as points in maps. The quality of the PCA model can be evaluated using cross-validation techniques such as the bootstrap and the jackknife. PCA can be generalized as *correspondence analysis* (CA) in order to handle qualitative variables and as *multiple factor analysis* (MFA) in order to handle heterogeneous sets of variables. Mathematically, PCA depends upon the eigen-decomposition of positive semi-definite matrices and upon the singular value decomposition (SVD) of rectangular matrices. © 2010 John Wiley & Sons, Inc. *WIREs Comp Stat* 2010 2 433–459

- Principal Component Analysis by I.T. Jolliffe, 2nd edition.



1 Motivation

1.1 Motivation

California Housing Prices Dataset

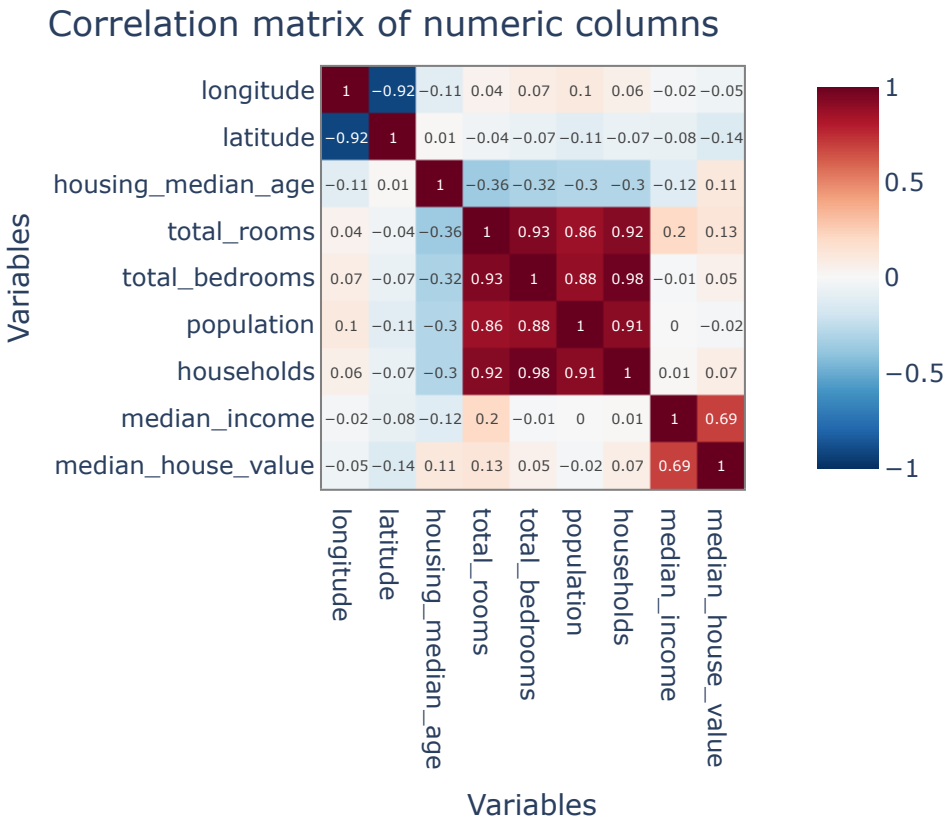
- The data contains information of neighborhoods from the 1990 California census.
- Dimension: (20640, 10).

	latitude	total_rooms	population	median_income	median_house_value	ocean_proximity
0	37.88	880.0	322.0	8.3252	452600.0	NEAR BAY
1	37.86	7099.0	2401.0	8.3014	358500.0	NEAR BAY
2	37.85	1467.0	496.0	7.2574	352100.0	NEAR BAY
3	37.85	1274.0	558.0	5.6431	341300.0	NEAR BAY
4	37.85	1627.0	565.0	3.8462	342200.0	NEAR BAY

- How are the `size` and `economic` of neighborhoods related?
- How are the previous related to their locations?

- Correlation matrix

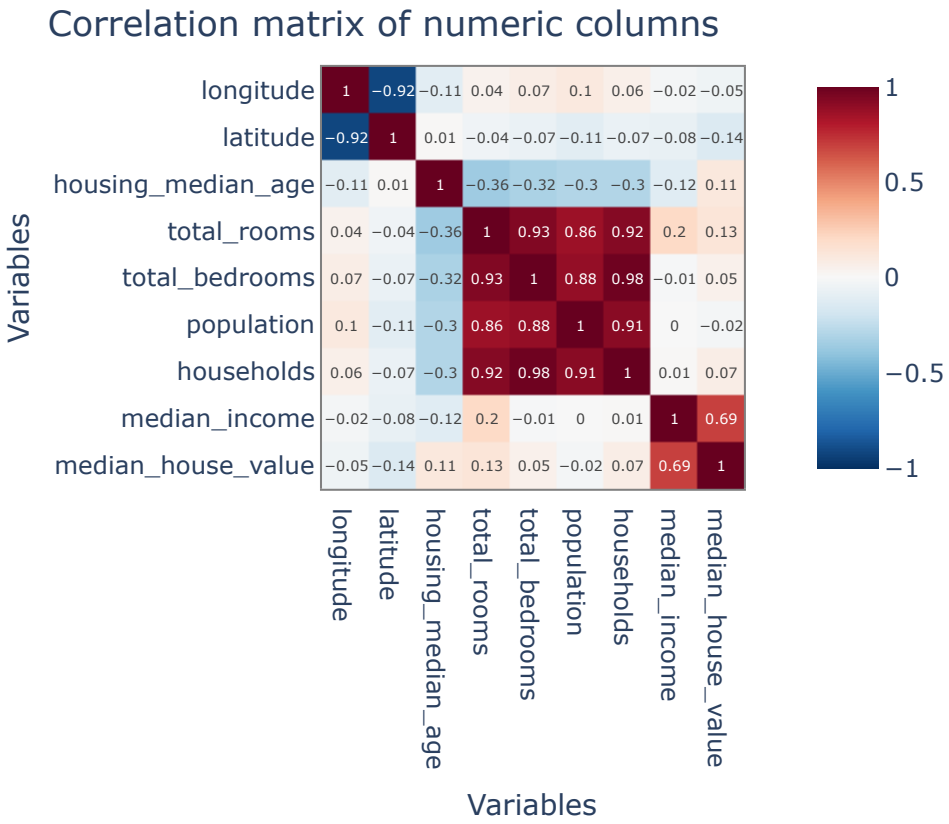
▶ Code



- **Recall:** correlation matrix:
  - Summarizes linear relation between columns.
  - Absolute values ≈ 1 indicate strong linear relation.
  - Weak values ≈ 0 suggest non-linear relation, or may be affected by
    - Outliers (see TP2 Solution),
    - Lacks of data points,
    - Confounding variables…

- Correlation matrix

▶ Code



- High correlations among columns suggest similar information therefore can be compressed / reduced.
- **Geometrically**, the pattern of data can be represented in simpler / smaller dimension.
- **PCA's Objectives**:
  - **Detect important directions** of data cloud
  - **Represent the data** in terms of these dimensions.

- **Application**: **Dimensionality reduction**, **data visualization** & **compression**, **feature clustering** & *extraction*, *noise reduction*, etc.

2 Principal Directions (PDs) & Components (PCs)

2.1 PDs & PCs

Principal Directions (PDs)

- Given data matrix: $\boldsymbol{D} \in \mathbb{R}^{n \times d}$,

	X	Y	Z
0	0.041	1.192	1.591
1	-1.439	2.870	1.992
2	1.661	2.043	4.787
3	-2.891	0.794	-1.043
4	1.618	3.229	3.470
5	2.310	2.082	3.454

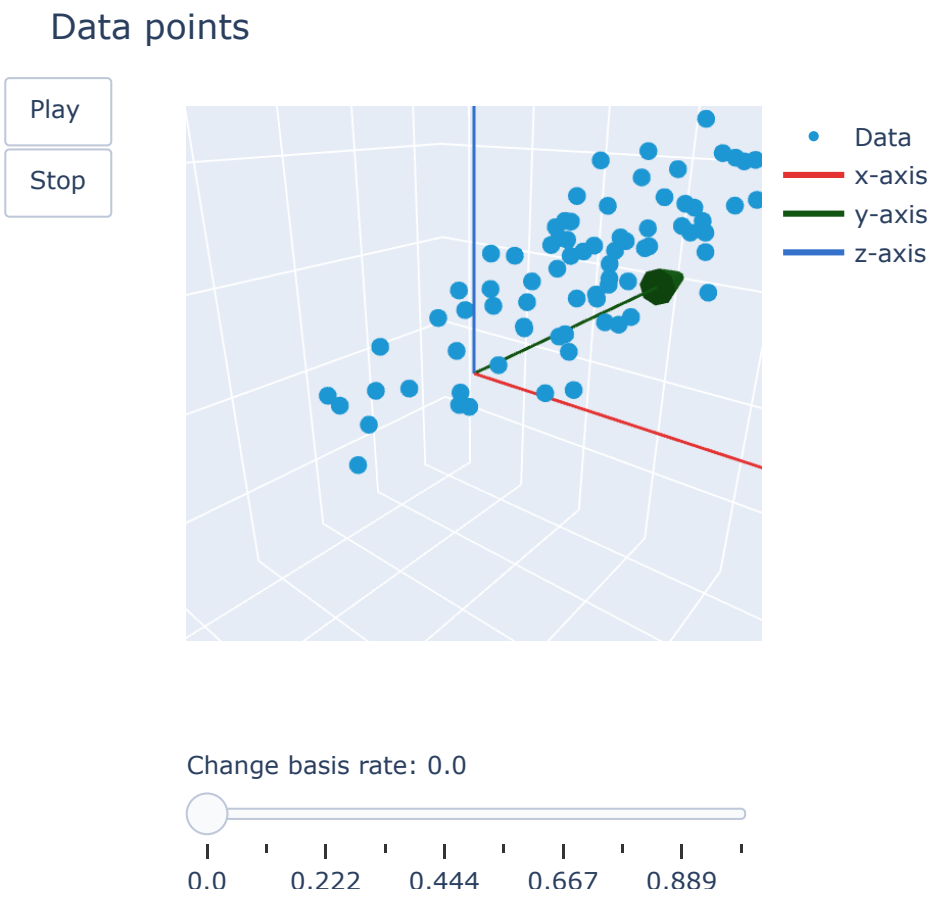


Principal Directions (PDs)

- **Principal Directions (PDs)** are the so-called **important directions** mentioned in **PCA's objectives**.

Definition: Principal Directions $\vec{u}_k$

- **The 1st PD** $\vec{u}_1 \in \mathbb{R}^d$ is the *unit vector or direction* that the data **spreads** or **varies** the most: $\vec{u}_1 = \arg\max_{\vec{u}\in\mathbb{R}^d:\|\vec{u}\|^2=1} \|D\vec{u}\|^2$.
- **The 2nd PD** $\vec{u}_2 \in \mathbb{R}^d$ is the *unit direction* **perpendicular** to $\vec{u}_1$ and captures **2nd largest variation** of the data: $\vec{u}_2 = \arg\max_{\vec{u}\in\mathbb{R}^d:\|\vec{u}\|^2=1,\vec{u}\perp\vec{u}_1} \|D\vec{u}\|^2$.
- **The k-th PD** $\vec{u}_k \in \mathbb{R}^d$ $(3 \le k \le d)$ is the *unit direction* that is **perpendicular** to $S_{k-1} = \{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_{k-1}\}$ and captures the k-th largest variation of the data: $\vec{u}_k = \arg\max_{\vec{u}\in\mathbb{R}^d:\|\vec{u}\|^2=1,\vec{u}\perp S_{k-1}} \|D\vec{u}\|^2$.



Principal Components (PCs)

- **Principal Directions (PDs)** are the so-called **important directions** mentioned in **PCA's objectives**.

Definition: Principal Components PC_j

- Let $\boldsymbol{U}$ be a $d \times d$ matrix with all **PDs** columns, i.e.,

$$\boldsymbol{U} = [\vec{u}_1 | \vec{u}_2 | \dots | \vec{u}_d],$$

  then $\tilde{\boldsymbol{D}} = \boldsymbol{D}\boldsymbol{U}$ is the **rotated data** with the following properties:

  - *Independent cols*: $\tilde{\boldsymbol{D}}_j \perp \tilde{\boldsymbol{D}}_k, \forall j \neq k$.
  - *For any* $j = 1, 2, \dots, d$: $\tilde{\boldsymbol{D}}_j$ is the column with j-th **largest variation**.
  - *Total var*: $\sum_{j=1}^{d} \|\tilde{\boldsymbol{D}}_j\|^2 = \sum_{j=1}^{d} \|\boldsymbol{D}_j\|^2$.

- The columns $\tilde{\boldsymbol{D}}_j$'s are called **Principal Components (PCs)** of $\boldsymbol{D}$ denoted by PC_j.



How To Find PDs & PCs

- Perform **PCA** comes down to computing rotated data $\tilde{D} = DU$ or computing U.

Principal Direction Problem

- Without loss of generality, we assume that the data is **centered** ($\overline{D}_j = 0, \forall j = 1, \ldots, d$).

1. The **1st PD** $\vec{u}_1$ is the **eigenvector** of matrix $A = D^T D$ with the **largest eigenvalue** $\lambda_1 \geq 0$, i.e., $(\vec{u}_1, \lambda_1)$ solves $A\vec{u}_1 = \lambda_1 \vec{u}_1$.
2. The **2nd PD** $\vec{u}_2$ is the **eigenvector** of matrix $A = D^T D$ with the **2nd largest eigenvalue** $\lambda_2 \geq 0$, i.e., $(\vec{u}_2, \lambda_2)$ solves $A\vec{u}_2 = \lambda_2 \vec{u}_2$.
3. The **k-th PD** $\vec{u}_k$ is the **eigenvector** of matrix $A = D^T D$ with the **k-th largest eigenvalue** $\lambda_k \geq 0$, i.e., $(\vec{u}_k, \lambda_k)$ solves $A\vec{u}_k = \lambda_k \vec{u}_k$.

Solution to Principal Directions $\vec{u}_k$

- **PDs** $\vec{u}_1, \ldots, \vec{u}_d$ are the **1st, ..., d-th eigenvectors** of matrix $A = D^T D$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$ respectively.
- Principal components are the columns of

$$\underbrace{\tilde{D}}_{n \times d} = \underbrace{D}_{n \times d} \times \underbrace{U}_{n \times d} = \begin{bmatrix} x_{11} & \cdots & x_{1d} \\ x_{21} & \cdots & x_{2d} \\ x_{31} & \cdots & x_{3d} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nd} \end{bmatrix} \begin{bmatrix} u_{11} & \cdots & u_{1d} \\ u_{21} & \cdots & u_{2d} \\ \vdots & \ddots & \vdots \\ u_{d1} & \cdots & u_{dd} \end{bmatrix}$$

- The PC_j is the combination of original columns of D_k's, i.e., $\text{PC}_j = \sum_{k=1}^{d} u_{kj} D_k$.

Implementation Of PCA

Unnormalized PCA

- Given data: $D \in \mathbb{R}^{n \times d}$
- Centering $D_c = \underbrace{D}_{n \times d} - \underbrace{\mathbb{1}_n}_{n \times n} \overbrace{D}^{n \times d} / n.$
- Perform **Eigenvalue Decomposition** on **covariance matrix** $V = D_c^T D_c/(n-1)$, i.e.,

$$V = U \Lambda U^T,$$

  - U: matrix of eigenvectors
  - $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$: diagonal matrix of eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$.

Normalized PCA

- Given data: $D \in \mathbb{R}^{n \times d}$
- Standardizing: compute std matrix $\Sigma = \mathrm{diag}(\hat{\sigma}_1, \ldots, \hat{\sigma}_d)$ and standardized data $Z = (\underbrace{D}_{n \times d} - \underbrace{\mathbb{1}_n}_{n \times n} \overbrace{D}^{n \times d} / n)\Sigma^{-1}$.
- Perform **Eigenvalue Decomposition** on **correlation matrix** $C = Z^T Z/(n-1)$, i.e.,

$$C = U \Lambda U^T.$$

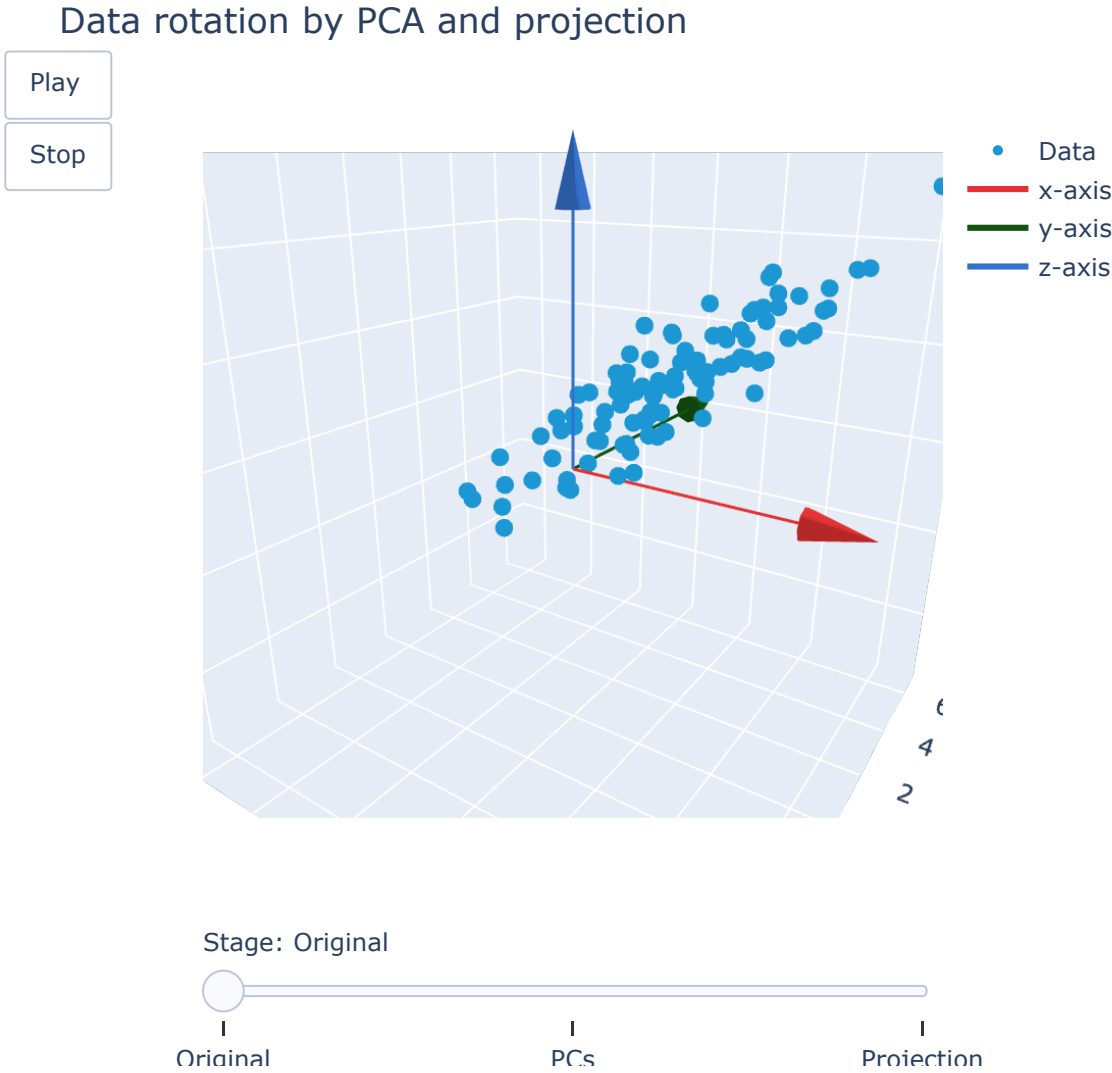
- Rotated data: $\tilde{D} = DU$. [👉 Demo]

3 Dimensional Reduction Reconstruction

3.1 Reduction/Reconstruction

Dimensional Reduction

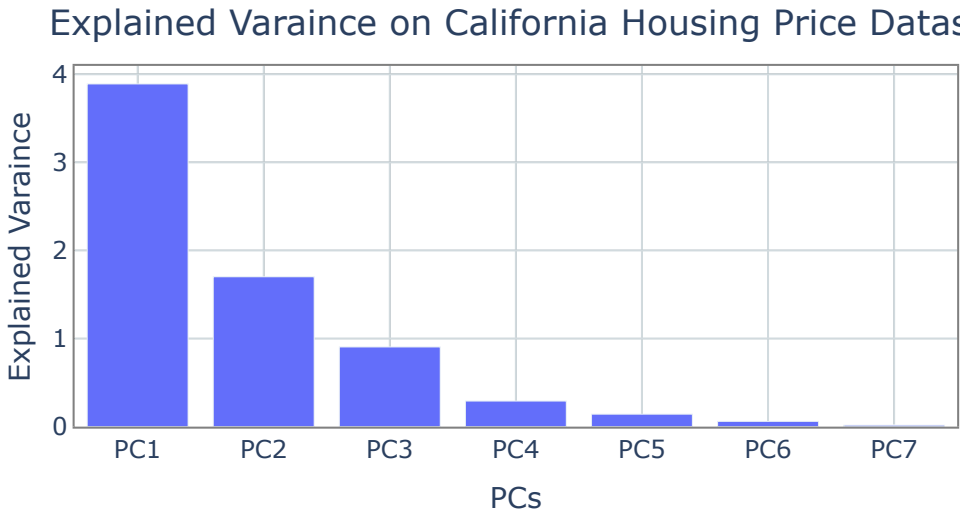
- **Reduction** to k dimensions ($k < d$) is done by $\tilde{D}_k = \tilde{D}[:, :k]$.
- **Quality of representation** on each direction: $\left[\frac{\lambda_1}{\sum_{j=1}^{d} \lambda_j}, \ldots, \frac{\lambda_k}{\sum_{j=1}^{d} \lambda_j}\right]$.
- **Variance explained** by k dimensions: $\frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{d} \lambda_j} \times 100$.
- **For visualization**, one can choose $k = 2$ or 3. [👉 Demo]



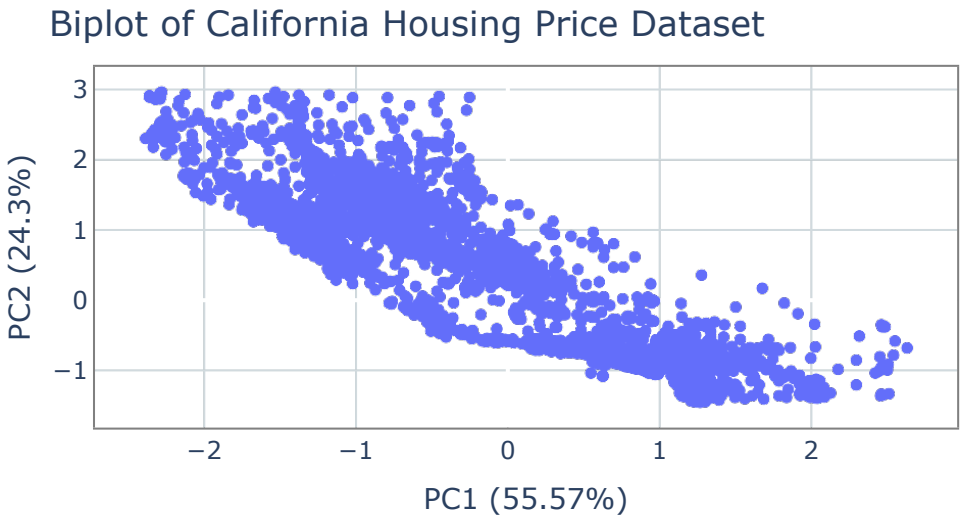
Dimensional Reduction

- **Reduction** to k dimensions ($k < d$) is done by $\tilde{D}_k = \tilde{D}[:, :k]$.
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- **For visualization**, one can choose $k = 2$ or 3. [👉 Demo]

- Application on California Housing:



- Variance explained by 2 PCs: 79.86%.



Dimensional Reduction

- **Reduction** to k dimensions ($k < d$) is done by $\tilde{D}_k = \tilde{D}[:, :k]$.
- **Quality of representation** on each direction: $\left[\frac{\lambda_1}{\sum_{j=1}^{d}\lambda_j}, \dots, \frac{\lambda_k}{\sum_{j=1}^{d}\lambda_j}\right]$.
- **Variance explained** by k dimensions: $\frac{\sum_{j=1}^{k}\lambda_j}{\sum_{j=1}^{d}\lambda_j} \times 100$.
- **For visualization**, one can choose $k = 2$ or 3. [👉 Demo]

Reconstruction

- The **reconsution** is defined by:

$$D_k^{\text{rec}} = \tilde{D}_k U[:, :k]^T.$$

- **Low-rank approximation** shows that this reconstruction D_k^{rec} is the best k-rank approximation of D w.r.t. **Frobenius norm**, i.e.,

$$D_k^{\text{rec}} = \arg \min_{M_k : r(M_k) \le k} \|X - M_k\|_F,$$

where $\|D\|_F = \sqrt{\sum_{i,j} D_{ij}^2}$ is the **Frobenius norm**.

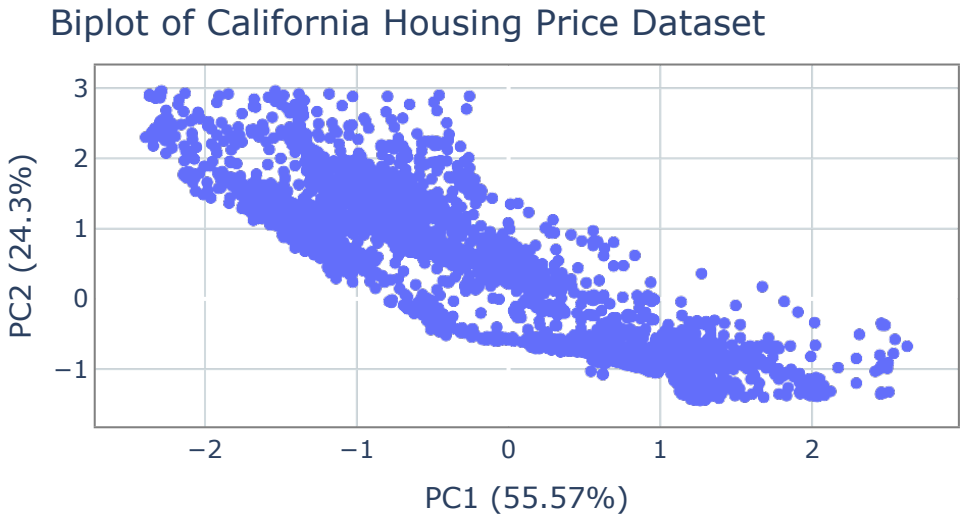
Summary

- PCA captures and arranges directions with largest, 2nd largest,… variation into the 1st, 2nd, … column using Spectral Theorem (Eigenvalue decomposition).
- These directions are contained in the eigenvector matrix of **Covariance** (unnormalized) or **Correlation** matrix (normalized) of the data.
- The variation along each direction is given by the corresponding eigenvalues: $\lambda_1 > \cdots > \lambda_d$.
- Our goal is to explore dimensional reduction and reconstruction power of PCA, however, it can be used in Data Analysis, EDA, Data Summary…
- **Remark:** Many highly correlated columns indicates effective dimensional reduction (the first two PCs can capture significant variation).

4 Correlation Circle

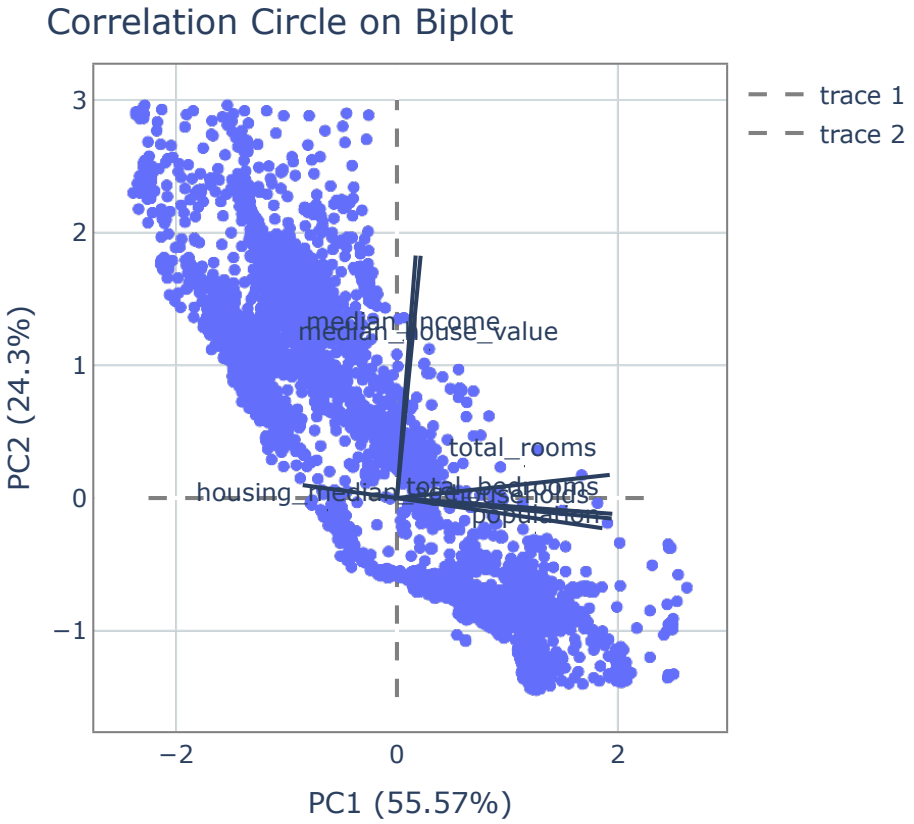
4.1 Correlation Circles

- Besides dimensional reduction, `PCA` can also be used to summarize data.

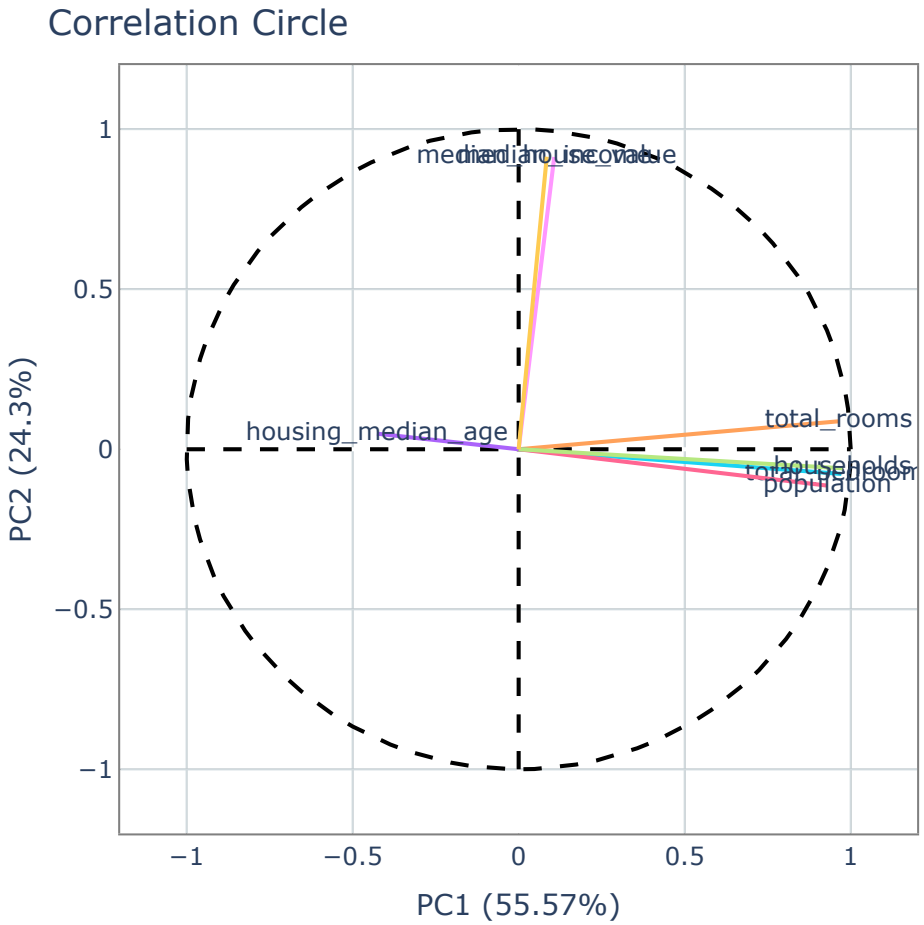


- How would you explain this graph?
- 🔑 The connection between PCs and the original columns.
- **Correlation Circle** is the key.

- One can compute correlations between original data with PCs:



Interpretation



- **Long** variables indicate **large** correlation with the PCs therefore well represented by the PCs.
- *Highly correlated* original variables are mostly aligned.
- Those that are *nearly perpendicular*, tend to be decorrelated.
- **PC1** is strong related to **neighborhood size** variables.
- **PC2** is strong reated to **financial condition** variables.
- Correlation circles can be used to summarize the **Biplot** of individuals.

- Correlation circles can summarize the **relationship between original variables**, **between original variables and PCs** and help interpreting the **Biplot**.
- ⚠ **Caution**: Interpretation of PCs may be misleading! It's intuition not the fact!

5 Contribution Scores

5.1 Variable Contributions

- From the **Correlation Circle**, we can compute the contribution of original variable D_j on PC_k by

$$\mathrm{Ctr}(D_j, \mathrm{PC}_k) = r_{jk}^2 / \lambda_k = \mathrm{corr}^2(D_j, \mathrm{PC}_k)/\lambda_k.$$

▶ Code

	PC1	PC2
housing_median_age	0.047	0.001
total_rooms	0.239	0.004
total_bedrooms	0.243	0.003
population	0.222	0.008
households	0.244	0.002
median_income	0.003	0.490

▶ Code

	PC1	PC2
median_house_value	0.002	0.491

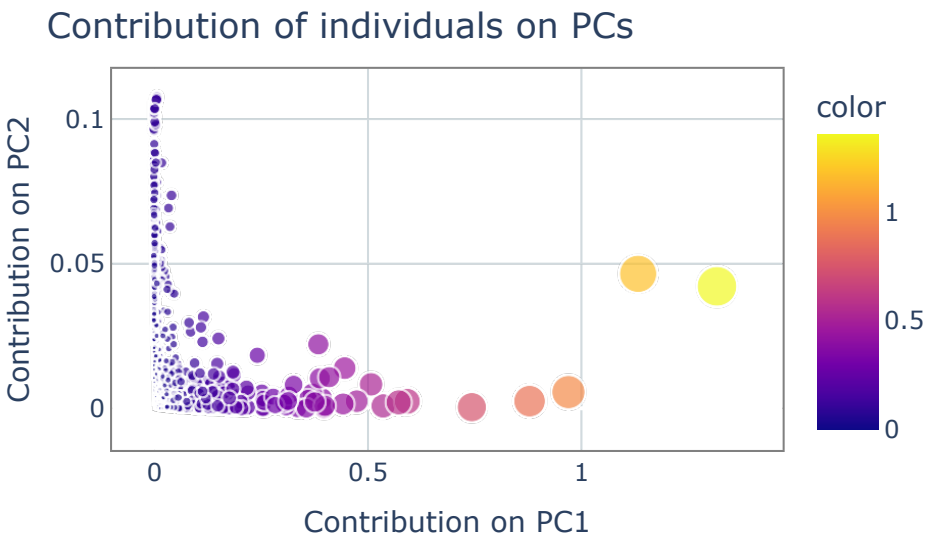
- The 4 **Neighborhood Size** variables contribute toally 94.84% of the total variation of PC1.
- The 2 **Financial Condition** variables contribute toally 98.09% of the total variation of PC2.

5.2 Observation Contributions

- **The contribution of individual i on PC k** is defined by

$$\text{Ctr}(x_i, PC_k) = \frac{[DU]_{ik}^2}{\sum_{\ell=1}^{n} [DU]_{\ell k}^2} = \frac{[DU]_{ik}^2}{(n-1)\lambda_k}$$

▶ Code

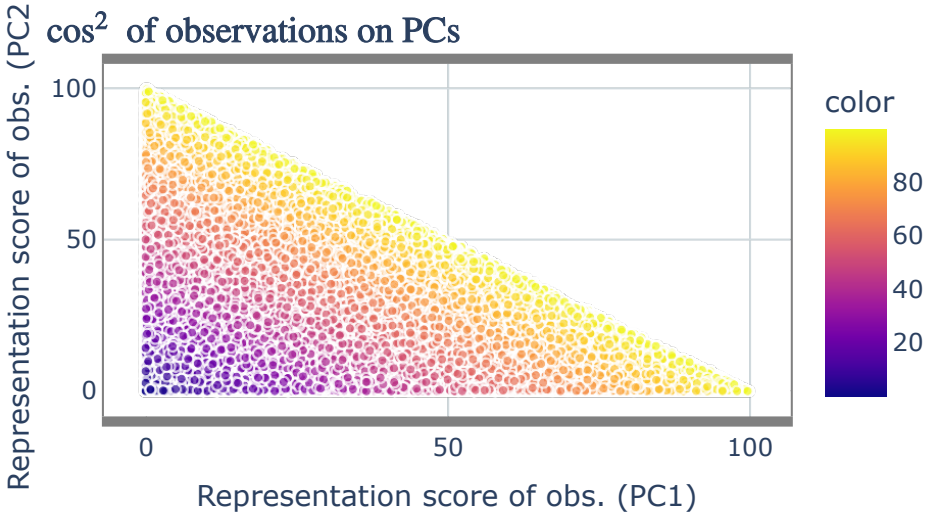


- Neighborhood 9782 has highest contribution of 1.32% on PC1, i.e., it may be the largest or smallest block.
- Neighborhood 16013 has highest contribution of 0.11% on PC2, i.e., it may be the wealthiest or poorest block.

5.3 Squared Consine Scores

- **Question:** Is the obs. i accurately represented by PC k?
- **Answer:** $\cos^2(\mathrm{x}_i, \mathrm{PC}_k) = \frac{z_{ik}^2}{\sum_{p=1}^{n} z_{ip}^2} = \frac{z_{ik}^2}{\|z_i\|^2}$, with $z_i = U^T \mathrm{x}_i = \begin{pmatrix} \vdots \\ z_{ik} \\ \vdots \end{pmatrix}$.

▶ Code



- Neighborhood 875 has highest $\cos^2 = 99.72\%$ with PC1, i.e., it's well represented/explained by PC1.
- Neighborhood 18091 has highest $\cos^2 = 99.61\%$ with PC2, i.e., it's best explained by PC2.

6 Variable Clustering

6.1 Variable Clustering

VARCLUS Algorithm (SAS)

1. **Start:** Perform PCA on all variables $C = [D_j]$ (1 cluster) to obtain eigenpairs: U, Σ. Let $d_C = \#C$ and $U_2 = U[:, :2]$.
2. **While** $\lambda_2 >$ MAXEIGEN **do**:

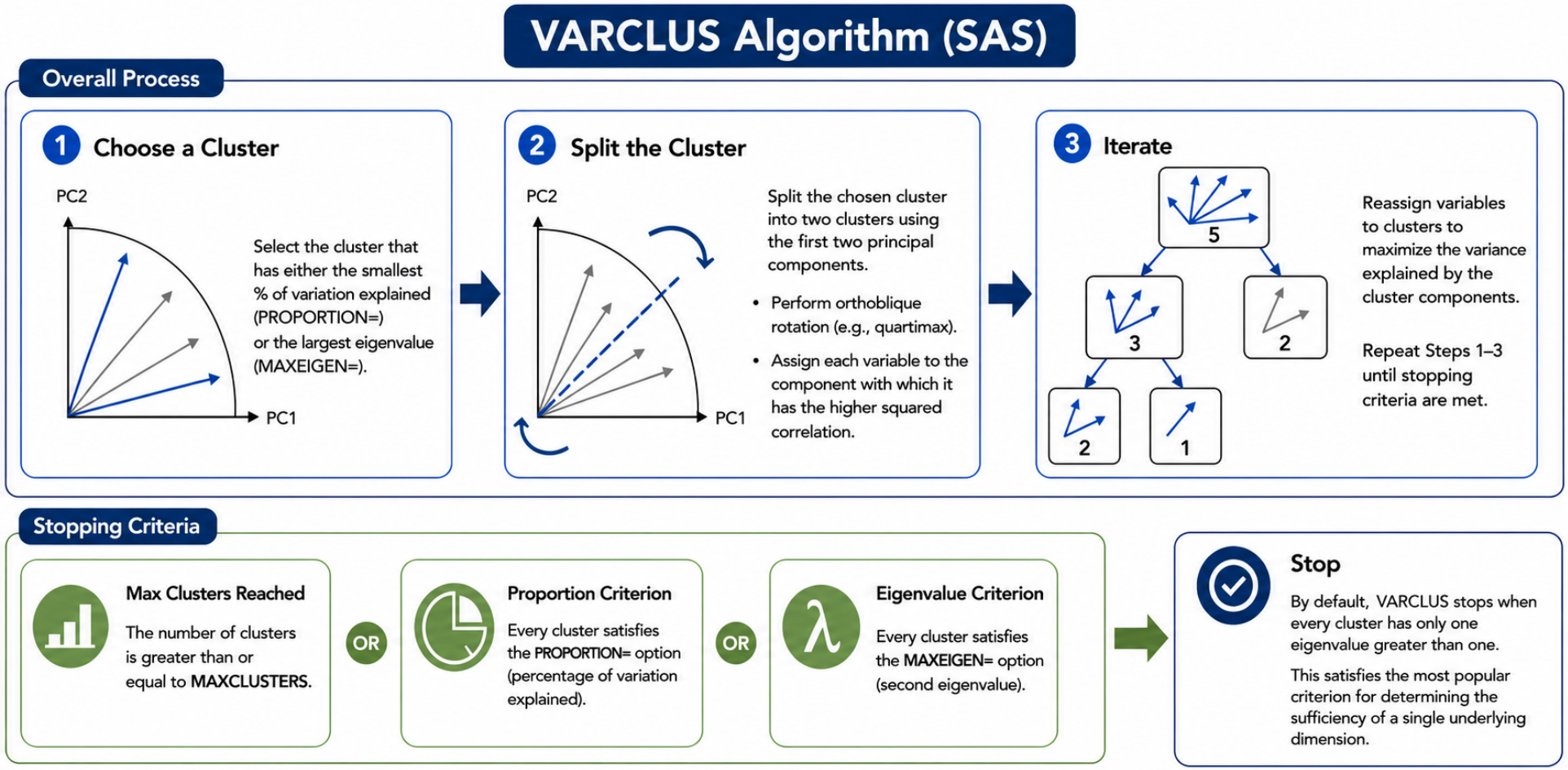
- **Orthoblique Rotatioin:** Compute rotation matrix $T \in \mathbb{R}^{2\times 2}$ defined by

$$T = \arg \max_{V: V^T V = I_2} \sum_{i=1}^{d_C} \sum_{j=1}^{2} [U_2 V]_{ij}^4.$$

- Compute the rotated data: $R = CT$.
- **Reassign variables to new cluster:** For any variable D_j:
  - If $r^2(D_j, R_1) > r^2(D_j, R_2)$, then $D_j \to C_1$,
  - else $D_j \to C_2$.

3. Set C to be each of the C_1 and C_2.
4. Repeat 1-3.

Note: Common choices of MAXEIGEN are 0.7 or 1.0



Summary of VARCLUS

- It can helps *mitigating severe multicollinearity, performing feature selection, enhacing model interpretability & deplolyment…*
- The performance may be different according to the stopping criterion.

Application

- `VARCLUS` on California Housing Price dataset with `MAXEIGEN = 0.7`.
- Here, the R-square scores are defined as:
  - $\text{RS_own}_k(X_j) = \text{corr}^2(X_j, \text{PC1}_k)$,
  - $\text{RS_NC}_k(X_j) = \max_{\ell \neq k}\{\text{corr}^2(X_j, \text{PC1}_\ell)\}$
  - Unexplained variance: $\textbf{RS_Ratio} = \frac{\text{RS_NC}}{\text{RS_own+RS_NC}}$.

▶ Code

	Cluster	Variable	RS_Own	RS_NC	RS_Ratio
0	0	total_rooms	0.919438	0.130053	9.260583e-02
1	0	total_bedrooms	0.961261	0.102689	4.317264e-02
2	0	population	0.886009	0.087490	1.249208e-01
3	0	households	0.970140	0.091668	3.287341e-02
4	1	median_income	0.844178	0.013990	1.580331e-01
5	1	median_house_value	0.844178	0.011328	1.576076e-01
6	2	housing_median_age	1.000000	0.109511	2.493512e-16



🥳 Yeahhhh….

Let’s Party… 🥂